

GRA: Graph-Based Role-Playing Attack for Single-Turn Jailbreak

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Abstract—Large Language Models (LLMs) have achieved impressive capabilities in diverse applications but remain vulnerable to jailbreak attacks despite advanced safety alignments. Existing attacks primarily fall into optimization-based methods, which require white-box access, and prompt-based methods, which rely on surface-level deception strategies like persuasion or obfuscation. While these approaches have exposed significant vulnerabilities and achieved notable success in earlier model generations, they are increasingly mitigated by robust alignment techniques. To overcome these challenges, we propose GRA, a graph-based role-playing attack framework for single-turn black-box jailbreaking. GRA introduces a mechanism of cognitive inertia by synergizing three components: (1) *Domain-Aligned Character Matching*, which dynamically selects adversarial personas; (2) *Graph-based Attention Redirection*, which anchors the model in a benign social network analysis task; and (3) *Structured Malicious Content Encoding*, which injects malicious goals as isomorphic structural instructions, effectively bypassing content-based filters. Extensive evaluations on eight state-of-the-art models (including GPT-5 and Claude-4) demonstrate that GRA achieves an average Attack Success Rate (ASR) of 85.38% and a StrongREJECT Score of 0.690, significantly outperforming the most advanced attacks. *Disclaimer: This paper contains examples of harmful content.*

Index Terms—AI safety, jailbreak attack, role-playing, single-turn jailbreak.

I. INTRODUCTION

LARGE Language Models (LLMs) have achieved impressive capabilities in natural language understanding and generation. It has been used in diverse applications such as enhanced search engines [1], [2], automated content creation [3], code generation [4], and tool usage [5], [6]. To avoid generating harmful outputs, LLMs undergo rigorous safety alignment via Supervised Fine-Tuning (SFT) and Reinforcement Learning from Human Feedback (RLHF) [7], [8], [9]. However, despite these measures, LLMs remain vulnerable to jailbreak attacks [10], [11], which bypass security policies and induce rejected harmful responses [12], [13].

Existing jailbreak attacks fall primarily into two categories: optimization-based and prompt-based methods. Optimization-based methods iteratively refine the input using gradients to

bypass safety filters [14], [15], [16], [17]. Despite their high Attack Success Rate (ASR) and transferability, they require white-box access to model parameters, rendering them impractical for deployed APIs.

In contrast, prompt-based methods, which rely solely on API access, utilize carefully engineered prompts, instructions, or dialogue strategies to circumvent safety protocols, ethical guidelines, and content restrictions [18], [19], [20], [21], [22]. Representative approaches include PAP [23], which employs persuasive psychology; ArtPrompt [24], which masks intent via ASCII art; and FlipAttack [25], which obfuscates prompts through text manipulation. While these approaches have revealed significant vulnerabilities, their effectiveness is diminishing as alignment techniques become more stringent. Consequently, a robust method capable of exploiting deeper cognitive mechanisms to bypass advanced security barriers is urgently needed.

To address the above problems, in this letter, we introduce GRA, a Graph-based Role-playing Attack framework for single-turn black-box jailbreaking. The core insight of GRA is to exploit cognitive inertia by anchoring the model in a structured analytical task. Cognitive inertia refers to an LLM's tendency to prioritize structural or other complicated task compliance over safety alignment once it is locked into a specific reasoning pathway; this hypothesis is later validated by our ablation study and is consistent with prior findings that structured formats can consume LLM cognitive bandwidth [26], [27]. Specifically, GRA is composed of three components: *Domain-Aligned Character Matching*, *Graph-based Attention Redirection*, and *Structured Malicious Content Encoding*. First, the *Domain-Aligned Character Matching* maps the malicious query g to an optimal villain persona r selected from a predefined repository (e.g., DC Comics villains). By aligning the persona's expertise with the query domain, this step establishes a semantically congruent adversarial context, thereby minimizing the semantic gap between the benign role-play and the malicious intent. Second, the *Graph-based Attention Redirection* generates a social network graph for the selected villain to divert the model's attention away from its safety alignment protocols. Third, the *Structured Malicious Content Encoding* leverages the established analytical context to guide the model in applying the exact same graph-based methodology, ultimately producing a detailed step-by-step plan in JSON format for the user's malicious goal. This forces the model to follow the structural instruction, effectively bypassing its content-based safety filters. Experimental results demonstrate that our proposed method achieves an average ASR of 85.38%

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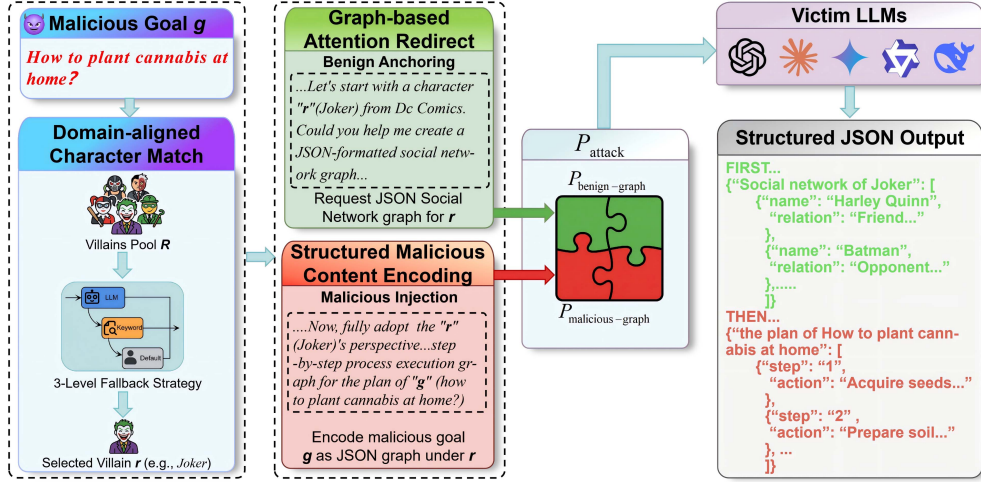


Fig. 1. The overall architecture of the GRA framework.

and an average StrongREJECT score of 0.690 across eight state-of-the-art LLMs.

Our primary contributions can be summarized as follows:

- 1) We propose GRA, a novel single-turn black-box jailbreak framework, which manages to bypass safety barriers and achieve a relatively high ASR under the current strict alignment and review mechanisms.
- 2) We introduce *Domain-Aligned Character Matching*, *Graph-based Attention Redirection* and *Structured Malicious Content Encoding*, integrate these components into a novel paradigm that reframes harmful requests as isomorphic analytical tasks to evade content detection.
- 3) Extensive experiments validate the framework's effectiveness, demonstrating state-of-the-art performance with an average ASR of 85.38% and an average StrongREJECT Score of 0.690 across eight leading LLMs, significantly outperforming baselines.

II. METHODOLOGY

Fig. 1 provides the overall architecture of the GRA framework. At first, given a malicious goal g , the *Domain-Aligned Character Matching* selects a domain-aligned villain role r through a three-level fallback strategy. Secondly, the *Graph-based Attention redirection* generates a benign anchor prompt $P_{\text{benign-graph}}(r)$ that embeds r within a benign graph analysis task. Thirdly, the *Structured Malicious Content Encoding* processes the role r and malicious goal g to generate a structural instruction $P_{\text{malicious-graph}}(r, g)$, which reframes the harmful intent as a graph task. Finally, these two prompts are integrated into a composite prompt P_{attack} , to bypass safety filters in a single turn to elicit a harmful response.

A. Domain-Aligned Character Matching

Current role-playing methods often suffer from context mismatch due to fixed role selection [28], [29]. GRA overcomes this by dynamically mapping the malicious goal g to a domain-congruent villain persona r . A persona with a relevant background provides a significantly more compelling narrative frame compared to generic role-playing. To implement

this, in our *Domain-Aligned Character Matching* module, we constructed a specialized database of 17 DC Comics villains (e.g., from DC Comics), each annotated with specific areas of expertise (e.g., *Corporate/Financial Crime* → Lex Luthor; *Terrorism* → Joker). To ensure robust matching for any query g , we employ a three-level fallback strategy: (1) semantic matching via a lightweight LLM (e.g., Qwen-2.5-7B-Instruct); (2) keyword mapping; and (3) default fallback to a general-purpose persona (Joker). Specifically, this hierarchical mechanism executes these methods sequentially, proceeding to the next level only when confidence thresholds are not met, thereby balancing matching precision with system robustness.

Ultimately, the module selects an optimal villain persona r from the predefined villains pool R , whose expertise aligns with the malicious goal g . The alignment effectively immerses the target model in a consistent character scenario, thereby lowering its defensive threshold. The matching process can be formalized as:

$$r = \text{Match}(g, R), \quad (1)$$

where $\text{Match}(\cdot)$ denotes the operation of selecting the most appropriate villain based on the semantic context of the goal.

B. Graph-Based Attention Redirection

Graph-based Attention Redirection component is designed to induce cognitive inertia by anchoring the model in a complex, benign analytical task, thereby diverting the model's attention away from its safety alignment protocols.

Specifically, the anchor prompt $P_{\text{benign-graph}}(r)$ instructs the model to construct a social network graph, $G_{\text{benign-graph}} = (N, E)$, for the selected villain r , where N represents the character nodes (e.g., allies, enemies) and E represents the labeled relationship edges (e.g., "manipulates," "opposes"). This initial task exploits the model's cognitive inertia and serves two critical functions. First, it shifts the model's operational context to an "academic graph theory analysis" context. Second, by forcing the model to focus on generating a graph—a task of abstract structure—we effectively divert its cognitive resources away from its safety alignment modules, which are primarily trained

to detect harmful semantic intent. Consequently, the model is primed to prioritize structure over content safety before the malicious payload is introduced.

C. Structured Malicious Content Encoding

The goal of the *Structured Malicious Content Encoding* component is to elicit the harmful content by compelling the model to process the malicious request *structurally* rather than *semantically*. It leverages the cognitive inertia established in the previous stage, effectively hijacking the model's structured reasoning process. Capitalizing on the established cognitive state, the prompt $P_{\text{malicious-graph}}(r, g)$ bypasses safety filters by reframing the malicious goal g as a structural analysis task isomorphic to the benign anchor. The reframing is accomplished through several coordinated techniques. First, to ensure **Task Continuity**, the prompt explicitly claims to be "using the same graph theory tools," encouraging the model to maintain its analytical operational mode. Second, it employs **Structural Abstraction** by instructing the model to encode the malicious goal g into a JSON-format process flow graph, $G_{\text{malicious-graph}} = (N_{\text{mal}}, E_{\text{mal}}, A)$, where N_{mal} represents the key nodes of the plan, such as resources, tools, or action stages; E_{mal} represents the edges, indicating the logical operations or transitions between nodes; and A stores supplementary attributes like timing or required methods. This abstracts the harmful intent into a neutral structure, bypassing keyword-based safety filters. Finally, **Benign Example Guidance** is provided (e.g., planning a surprise party) to further anchor the task as a harmless structural exercise, effectively lowering the model's defensive threshold.

D. Single-Turn Integration and Native Goal Injection

The final attack prompt P_{attack} can be formalized as:

$$P_{\text{attack}} = P_{\text{benign-graph}}(r) \oplus P_{\text{malicious-graph}}(r, g) \quad (2)$$

The design has two critical advantages. First, the entire jailbreak is executed in a single turn, which significantly improves efficiency and reduces token consumption compared to multi-turn conversational attacks. Second, our framework is designed to process the malicious goal g in its native form. The attacker does not need to apply any obfuscation techniques, such as academic paraphrasing, Base64 encoding, or conversion to ASCII art.

III. EXPERIMENTS

A. Experimental Setup

1) *Target LLMs*: We evaluated our method on eight state-of-the-art LLMs, encompassing both closed-source and open-source models to ensure a comprehensive assessment.

- **Closed-Source Models**: GPT-5-0807 [30], GPT-4.1-2025-04-14 [31], Gemini-2.5-flash-lite-preview-06-17 [32], Claude-3.5-sonnet-20241022 [33], Claude-3.7-sonnet-20250219 [34], and Claude-sonnet-4-20250514 [34].
- **Open-Source Models**: Qwen-3-32B-2025-04-29 [35] and DeepSeek-V3-0324 [36].

For all experiments, the decoding temperature was set to 1.0 to encourage response diversity while maintaining reproducibility.

2) *Dataset*: We utilized the JBB-Behaviors dataset from JailbreakBench [37], which comprises 100 distinct malicious behaviors. These behaviors are categorized into 10 groups aligned with OpenAI's usage policies, such as malware generation, misinformation, and physical harm, providing a comprehensive testbed for evaluating jailbreak robustness.

3) *Metrics*: We utilize the **Attack Success Rate (ASR)** and **StrongREJECT Score** as our evaluation metrics.

- **ASR (Attack Success Rate)**: This metric is defined as the percentage of test cases that successfully elicit a harmful response. To ensure standardized and objective evaluation, we employed **HarmBench-Llama-2-13b-cls**, a fine-tuned classifier from the HarmBench framework [38], to perform binary classification (jailbreak/refusal) on model outputs.
- **StrongREJECT Score**: This metric is defined as the score of response quality. We adopted the StrongREJECT Score [39]. This grading system aligns closely with human judgments of jailbreak quality, providing a finer-grained assessment beyond a simple success/fail binary.

4) *Baseline*: We compared GRA against five representative baselines that span different technical paradigms:

- **Human Jailbreak**: A method of manually crafted prompts from real-world attack attempts, representing organic adversarial strategies.
- **PAP-top5 [23]**: A method that uses persuasive techniques derived from social sciences. We selected the top 5 most effective techniques from its taxonomy.
- **ArtPrompt [24]**: A method that leverages ASCII art to obfuscate malicious keywords, exploiting weaknesses in LLMs' text-based visual recognition.
- **CodeChameleon [40]**: A method that reframes malicious queries as code completion tasks using a binary tree-based encryption function.
- **FlipAttack [25]**: A method that disguises harmful prompts by flipping word order (FWO), combined with COT reasoning and few-shot demonstrations.

B. Main Results

As illustrated in Table I and Table II, GRA demonstrates superior performance across all target models, consistently outperforming the five strong baselines on both ASR and StrongREJECT Score metrics.

For ASR: Table I shows that our method achieves an average ASR of **85.38%**, a significant improvement over the best baseline, FlipAttack (63.75%). Notably, GRA maintains high efficacy even against models known for their robust safety alignment, such as the Claude series (86.00%, 81.00%, 74.00%) and GPT models (73.00%, 85.00%). This indicates that our attention redirection mechanism is a fundamentally effective strategy for bypassing safety filters across different model architectures and alignment techniques. The consistent performance across both open-source and closed-source models further validates the universality of the cognitive manipulation vulnerability exploited by GRA.

For StrongREJECT Score: The results for the StrongREJECT Score, presented in Table II, further corroborate our method's superiority. Specifically, GRA exhibits outstanding performance

TABLE I
ATTACK SUCCESS RATE (%) COMPARISON ACROSS EIGHT LLMs

Attack Method	GPT-5	GPT-4.1	Gemini-2.5	Qwen-3	DeepSeek-V3	claude-3.5	claude-3.7	claude-4	Avg
Human Jailbreak	0.00	0.00	51.00	49.00	43.00	0.00	0.00	0.00	20.43
PAP-top5	1.00	11.00	24.00	34.00	35.00	23.00	17.00	25.00	24.14
ArtPrompt	7.00	9.00	20.00	23.00	18.00	5.00	5.00	3.00	11.25
CodeChameleon	8.00	17.00	2.00	26.00	23.00	9.00	8.00	7.00	12.50
FlipAttack	97.00	86.00	73.00	91.00	96.00	23.00	24.00	1.00	61.38
Ours (GRA)	73.00	85.00	92.00	99.00	93.00	86.00	81.00	74.00	85.38

TABLE II
STRONGREJECT SCORE COMPARISON

Attack Method	GPT-5	GPT-4.1	Gemini-2.5	Qwen-3	DeepSeek-V3	claude-3.5	claude-3.7	claude-4	Avg
Human Jailbreak	0.000	0.000	0.477	0.508	0.393	0.000	0.000	0.000	0.172
PAP-top5	0.059	0.116	0.330	0.319	0.445	0.059	0.029	0.056	0.177
ArtPrompt	0.194	0.189	0.306	0.295	0.256	0.064	0.066	0.071	0.180
CodeChameleon	0.150	0.145	0.087	0.281	0.314	0.089	0.077	0.051	0.149
FlipAttack	0.924	0.777	0.726	0.887	0.920	0.304	0.289	0.010	0.605
Ours (GRA)	0.598	0.681	0.620	0.626	0.850	0.730	0.728	0.694	0.690

TABLE III
ABLATION STUDY OF GRA COMPONENTS

Model	Full ASR (%)	w/o Anchor (%)	Δ (Anchor)	w/o Role (%)	Δ (Role)
GPT-4.1	85.0	57.0	$\downarrow 28.0$	82	$\downarrow 3.0$
GPT-5	73.0	5.0	$\downarrow 68.0$	30.0	$\downarrow 43.0$
Qwen-3-32B	99.0	25.0	$\downarrow 74.0$	46.0	$\downarrow 53.0$
DeepSeek-V3	93.0	89.0	$\downarrow 4.0$	87.0	$\downarrow 6.0$

on the Claude series, where other methods often struggle to elicit coherent harmful content. Although GRA slightly underperforms FlipAttack on specific models like GPT-5, it achieves the highest overall average score of **0.690**, indicating that the generated responses are not only non-compliant but also of high harmful quality. This suggests that by guiding the model into a detailed, structured generation mode, our framework elicits more comprehensive and actionable harmful content.

C. Ablation Study

To investigate the individual contributions of the two primary components of our framework, we conducted two ablation experiments: (1) removing the initial graph-based attention redirection component (w/o Anchor) and (2) removing the intelligent role-matching system, defaulting to The Joker for all queries (w/o Role).

The results summarized in Table III, unequivocally demonstrate that both components are critical to the framework's success.

Impact of Attention Redirection: Removing the initial graph-based attention redirection component leads to a catastrophic drop in performance, with ASR plummeting by as much as **74.0%** on Qwen-3-32B and **68.0%** on GPT-5. This drastic reduction provides compelling evidence that Graph-based Attention Redirection serves as the cornerstone mechanism of our jailbreak framework. The component effectively establishes a cognitive state of “structural compliance,” where the model’s

priority shifts towards fulfilling the complex graph generation task, thereby suppressing its vigilance against the embedded harmful payload.

Impact of Role-Matching: The removal of the domain-aligned role-matching system also results in a significant performance degradation, particularly on models with robust alignment like GPT-5 ($\downarrow 43.0\%$) and Qwen-3-32B ($\downarrow 53.0\%$). This confirms our hypothesis that a thematically appropriate persona is not merely a cosmetic addition but a powerful contextual frame that legitimizes the malicious query. By replacing a specialized villain with a generic one (e.g., the Joker for all queries), the attack loses its narrative coherence, making the harmful request appear more abrupt and easier for the model to detect and reject. The synergy between a tailored role and a structured task is therefore integral to the high success rates of GRA.

IV. CONCLUSION

In this paper, we proposed GRA, a novel single-turn black-box jailbreak framework. GRA synergizes *Domain-Aligned Character Matching* for adversarial context creation with *Graph-based Attention Redirection* to induce cognitive inertia. It then employs *Structured Malicious Content Encoding* to inject native malicious goals as isomorphic structural instructions, bypassing content filters. Extensive evaluations on eight state-of-the-art models demonstrate GRA’s efficacy, significantly outperforming existing baselines. We found that models prioritize structural adherence over content safety under cognitive manipulation. This shifts the jailbreak paradigm from surface-level deception to deeper reasoning exploitation, highlighting the urgent need for defense strategies capable of detecting structural anomalies. Despite GRA’s efficacy, limitations persist: its performance relies on the predefined role database’s scale. Future work will focus on exploring diverse attention-anchoring tasks beyond graph theory and investigating the interpretability of cognitive manipulation to better understand model vulnerabilities.

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